

BEE 4750 Homework 4: Linear Programming and Capacity Expansion

Name:

ID:

Due Date

Thursday, 11/06/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to analytically (or graphically) solve a linear program.
- Problem 2 asks you to formulate and solve a resource allocation problem using linear programming.
- Problem 3 asks you to formulate, solve, and analyze a standard generating capacity expansion problem.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```

Activating project at `c:\Users\nicho\OneDrive\Documents\hw4-naz-hw4`
Warning: The project dependencies or compat requirements have changed since the manifest wa
It is recommended to `Pkg.resolve()` or consider `Pkg.update()` if necessary.
@ Pkg.API C:\Users\nicho\.julia\juliaup\julia-1.12.1+0.x64.w64.mingw32\share\julia\stdlib\v
Precompiling packages...
      MbedTLS
      HTTP
0 dependencies successfully precompiled in 6 seconds. 219 already precompiled.

```

```

using JuMP
using HiGHS
using DataFrames
using Plots
using Measures
using CSV
using MarkdownTables

```

Problem 1:

Given the given constraints, we can identify some of the bounded intersection points.

- 1) At the minimums of (0, 1), this fits both inequalities, and has a max value of 2.
- 2) At (0, 4), this fits both inequalities, and has a max value of 8.

Then we can rearrange $x - 2y \leq -1$ to be $x \leq 2y - 1$ From here, we can substitute the minimum y value of $y = 1$, which, when plugged into the rearranged inequality, gives $x = 1$

- 3) From this, we can use (1, 1), this fits both inequalities, and has a max value of 5

Now we can try to maximize x with $x = 4$ for both inequalities

- 4) When $x = 4$ for the first inequality, this gives (4,3), which fits both inequalities and ranges, and has a max value of 18.
- 5) When $x = 4$ for the second inequality, $y = 2.5$, which fits both constraints, but has a max value of 17.

We can also try to find where both inequalities intersect by substituting $x = 2y - 1$ into the first inequality. Solving this gives a value of $17/6$ for y and $28/6$ for x or (4.67, 2.83) this, however, breaks the range for x so it is not feasible.

Therefore, the linear problem is maximized with the coordinates (4, 3) As it maximizes the $3x + 2y$ equation while fitting both the inequalities and the ranges.

Problems (Total: 30 Points)

Problem 1 (Total: 3 Points)

Solve the following linear program **analytically**. Show your work and justify any reasoning.

$$\begin{aligned} \max_{x,y} \quad & 3x + 2y \\ \text{subject to:} \quad & x + 4y \leq 16 \\ & x - 2y \leq -1 \\ & 0 \leq x \leq 4 \\ & 1 \leq y \leq 4. \end{aligned}$$

Problem 2 (12 points)

A farmer has access to a pesticide which can be used on corn, soybeans, and wheat fields, and costs \$70/kg-yr to apply. The crop yields the farmer can obtain following crop yields by applying varying rates of pesticides to the field are shown in Table 1.

Application Rate (kg/ha)	Soybean (kg/ha)	Wheat (kg/ha)	Corn (kg/ha)
0	2900	3500	5900
1	3800	4100	6700
2	4400	4200	7900

Table 1: Crop yields from applying varying pesticide rates for Problem 1.

The costs of production, *excluding pesticides*, for each crop, and selling prices, are shown in Table 2.

Crop	Production Cost (\$/ha-yr)	Selling Price (\$/kg)
Soybeans	350	0.36
Wheat	280	0.27
Corn	390	0.22

Table 2: Costs of crop production, excluding pesticides, and selling prices for each crop.

Recently, environmental authorities have declared that farms cannot have an *average* application rate on soybeans, wheat, and corn which exceeds 0.8, 0.7, and 0.6 kg/ha, respectively. The farmer has asked you for advice on how they should plant crops and apply pesticides to maximize profits over 130 total ha while remaining in regulatory compliance if demand for each crop (which is the maximum the market would buy) this year is 250,000 kg?

Problem 2.1

Formulate a linear program for this resource allocation problem, including clear definitions of decision variable(s) (including units), objective function(s), and constraint(s) (make sure to explain functions and constraints with any needed derivations and explanations). **Tip: Make sure that all of your constraints are linear.**

Problem 2.2

How many ha should the farmer dedicate to each crop and with what pesticide application rate(s)? How much profit will the farmer expect to make?

Problem 2.3

The farmer has an opportunity to buy an extra 10 ha of land. How much extra profit would this land be worth to the farmer? Discuss why this value makes sense and under what conditions you would recommend the farmer should make the purchase.

Problem 2 (15 points)

For this problem, we will use hourly load (demand) data from 2013 in New York's Zone C (which includes Ithaca). The load data is loaded and plotted below in Figure 1.

```
#Problem 2

crop_model = Model(HiGHS.Optimizer) #initialize model object

#establish decision variables
#decision variables will be the area treated at each pesticide rate.
#3x3 for the 3 different crops and the 3 different application rates.
#the units for this will be hectares (Ha)

@variable(crop_model, A[1:3, 1:3] >= 0)

#Here we establish parameters for objective and final cost calculations

#matrix to store crop yield values in kg/ha
crop_yields = [
2900 3800 4400
3500 4100 4200
5900 6700 7900
```

```

]

prices = [0.36, 0.27, 0.22] #$/kg
prod = [350, 280, 390] #$/ha
pest_cost = [0, 70, 140] # $/ha
compliance = [0.8, 0.7, 0.6] #kg/ha
demand = [250000, 250000, 250000] # $
land = 130 #ha

#I dropped the time component from some of
#the units since we're bounded to a one year calculation
#and the units are per year, so it naturally cancels out

#establish objective function

#here the farmer wants to maximize their profits while
#complying with regulatory compliance and the demand for each crop.

#So we need to account for the selling price * crop yield while also
#considering the production cost for each crop and the cost for the pesticide rate.
#then multiplying this by the application rate matrix will generate
#specific values for each combination

@objective(crop_model, Max, sum((prices[i]*crop_yields[i, j] - prod[i]
- pest_cost[j]) * A[i, j] for i = 1:3, j = 1:3))

#establish constraints

#cannot exceed the farmers 130 total Ha, so the sum of the plots for
#the 3 crops and their 3 application rates must be less than 130 ha.

@constraint(crop_model, total_land, sum(A[i, j] for i = 1:3, j = 1:3) <= land)

#crop yields cannot exceed the given maximum the market would buy
#so the sum of crop yields across the production matrix must be less
#than the provided demand for each crop (which is why we use J for the
#specific crop row)

for i = 1:3

```

```

    @constraint(crop_model, sum(crop_yields[i,j] * A[i,j] for j = 1:3) <= demand[i])
end

#constraint for compliance, based on application rate and application matrix
#Here we do j-1 to disregard the 0 pesticide application rate in the first position
#then sum the application rates 1 and 2 for each crop and make sure that it
#is less than the compliance multiplied by each portion of the production matrix
for i = 1:3
    @constraint(crop_model, sum((j-1)*A[i,j] for j = 1:3)
        <= (compliance[i] * sum(A[i, j] for j = 1:3)))
end

#optimize model code
optimize!(crop_model)

A_matrix = value.(A)

#print out results code
println("Optimal area (ha) per crop and pesticide rate:")
for i = 1:3
    for j = 1:3
        println("Crop $i, Rate $(j-1) kg/ha: $(A_matrix[i,j]) ha")
    end
end

for i = 1:3
    total_crop_area = sum(A_matrix[i,j] for j = 1:3)
    println("Total area for crop $i: $total_crop_area ha")
end

#Total profit summation code
profit = sum((prices[i]*crop_yields[i,j] - prod[i] - pest_cost[j]) * A_matrix[i,j]
    for i = 1:3, j = 1:3)

println("Expected total profit: $(round(profit, digits=2))")

```

Running HiGHS 1.11.0 (git hash: 364c83a51e): Copyright (c) 2025 HiGHS under MIT licence terms

LP has 7 rows; 9 cols; 27 nonzeros

Coefficient ranges:

Matrix [2e-01, 8e+03]

Cost [7e+02, 1e+03]

Bound [0e+00, 0e+00]

```

RHS      [1e+02, 2e+05]
Presolving model
7 rows, 9 cols, 27 nonzeros  0s
7 rows, 9 cols, 27 nonzeros  0s
Presolve : Reductions: rows 7(-0); columns 9(-0); elements 27(-0) - Not reduced
Problem not reduced by presolve: solving the LP
Using EKK dual simplex solver - serial
  Iteration      Objective      Infeasibilities num(sum)
        0      -1.8170298458e+02 Ph1: 7(28.3601); Du: 9(181.703) 0s
        7       1.1674116702e+05 Pr: 0(0) 0s
Model status      : Optimal
Simplex iterations: 7
Objective value    : 1.1674116702e+05
P-D objective error : 0.0000000000e+00
HiGHS run time     : 0.00
Optimal area (ha) per crop and pesticide rate:
Crop 1, Rate 0 kg/ha: 13.812154696132593 ha
Crop 1, Rate 1 kg/ha: 55.24861878453038 ha
Crop 1, Rate 2 kg/ha: 0.0 ha
Crop 2, Rate 0 kg/ha: 6.743306417339564 ha
Crop 2, Rate 1 kg/ha: 15.73438164045898 ha
Crop 2, Rate 2 kg/ha: 0.0 ha
Crop 3, Rate 0 kg/ha: 26.923076923076927 ha
Crop 3, Rate 1 kg/ha: 0.0 ha
Crop 3, Rate 2 kg/ha: 11.538461538461554 ha
Total area for crop 1: 69.06077348066297 ha
Total area for crop 2: 22.477688057798545 ha
Total area for crop 3: 38.46153846153848 ha
Expected total profit: 116741.17

```

Problem 2.2

The farmer should dedicate approximately 69.06 Ha to soybeans, 22.48 Ha to wheat, and 38.46 Ha to corn.

Additionally, the farmer should dedicate approximately 55.25 kilograms of pesticides for use on soybeans at application rate 1 kg/ha, 15.73 kilograms of pesticides to wheat at application rate 1 kg/ha, and 23.08 kilograms of pesticides to corn at application rate 2 kg/ha.

The farmer should expect a total profit of 116,741.17 dollars.

Problem 2.3

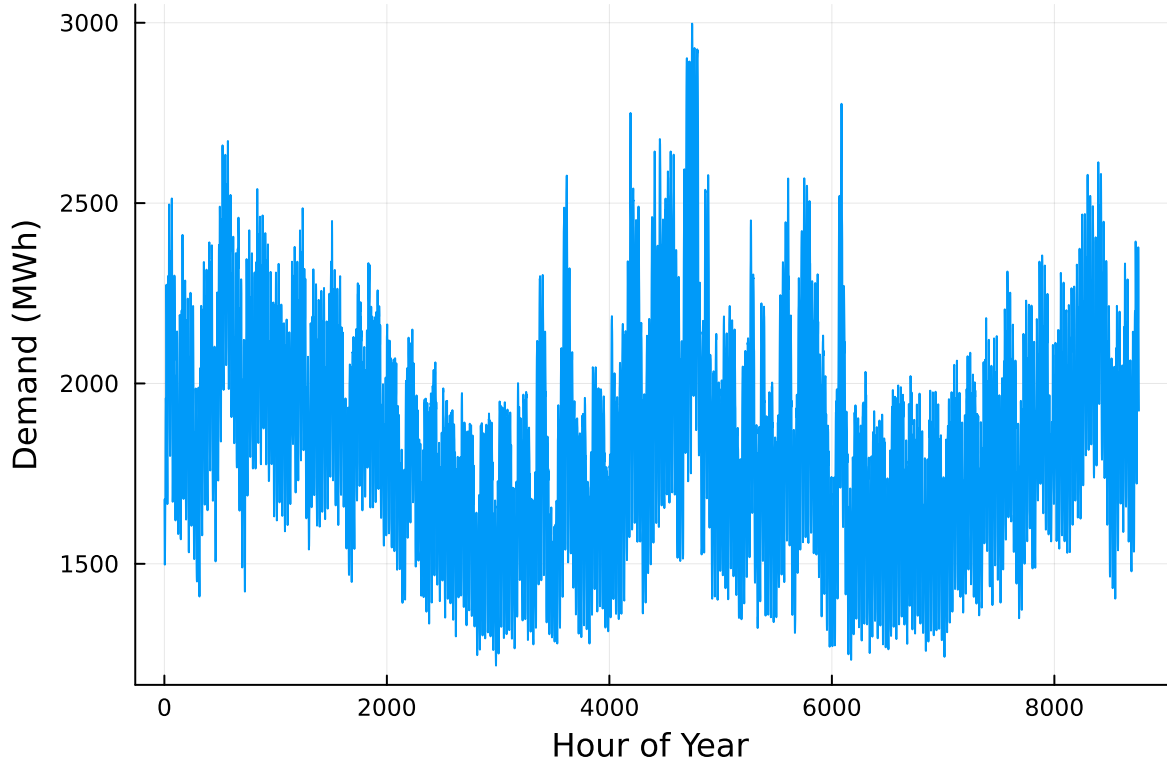
If the farmer were to buy an extra 10 ha of land, they would make 7,294 dollars in additional profit. This value can also be referred to as the shadow price of land, or how the solution would

improve if the constraint capacity was relaxed by one unit. In this case, how our objective function, profit, improves if land was increased. This value makes sense as we know the selling price for each of the crops as well as the production costs and the pesticide costs.

I would recommend the farmer to make the purchase if he thought that it would generate a positive return on investment. If he anticipates that land prices will continue to increase, then it is a good investment regardless of the additional annual profit. Additionally, while the land prices continue to increase year after year, he can generate an extra profit of 7,294 dollars per year. If we are solely looking at this through a one-year lense then I would recommend the farmer to purchase the land only if it was less than 7,294 dollars. However, I think it's more practical and logical to think about this over a longer period of time, even in a relatively short-term ownership of 10 years this would generate an additional 72,940 dollars of profit so if the land cost 50,000\$ the farmer would still generate over 20,000\$ of profit over 10 years as well as whatever land appreciation has occurred in the past decade.

```
# load the data, pull Zone C, and reformat the DataFrame
NY_demand = DataFrame(CSV.File("data/2013_hourly_load_NY.csv"))
rename!(NY_demand, "Time Stamp" => :Date)
demand = NY_demand[:, [:Date, :C]]
rename!(demand, :C => :Demand)
demand[:, :Hour] = 1:nrow(demand)

# plot demand
plot(demand.Hour, demand.Demand, xlabel="Hour of Year", ylabel="Demand (MWh)", label=:false)
```

Next, we load the generator data, shown in Table 3. This data includes fixed costs (\$/MW installed), variable costs (\$/MWh generated), and CO2 emissions intensity (tCO2/MWh generated).

```
gens = DataFrame(CSV.File("data/generators.csv"))
```

	Plant	FixedCost	VarCost	Emissions
	String15	Int64	Int64	Float64
1	Geothermal	450000	0	0.0
2	Coal	220000	24	1.0
3	NG CCGT	82000	30	0.43
4	NG CT	65000	40	0.55
5	Wind	91000	0	0.0
6	Solar	70000	0	0.0

Finally, we load the hourly solar and wind capacity factors, which are plotted in Figure 2. These tell us the fraction of installed capacity which is expected to be available in a given hour for generation (typically based on the average meteorology).

```

# load capacity factors into a DataFrame
cap_factor = DataFrame(CSV.File("data/wind_solar_capacity_factors.csv"))

# plot January capacity factors
p1 = plot(cap_factor.Wind[1:(24*31)], label="Wind")
plot!(cap_factor.Solar[1:(24*31)], label="Solar")
xaxis!("Hour of the Month")
yaxis!("Capacity Factor")

p2 = plot(cap_factor.Wind[4344:4344+(24*31)], label="Wind")
plot!(cap_factor.Solar[4344:4344+(24*31)], label="Solar")
xaxis!("Hour of the Month")
yaxis!("Capacity Factor")

display(p1)
display(p2)

```

You have been asked to develop a generating capacity expansion plan for the utility in Riley County, NY, which currently has no existing electrical generation infrastructure. The utility can build any of the following plant types: geothermal, coal, natural gas combined cycle gas turbine (CCGT), natural gas combustion turbine (CT), solar, and wind. The NY state legislature is planning to enact an annual CO₂ limit, which for the utility would limit the emissions in its footprint to 1.5 MtCO₂/yr.

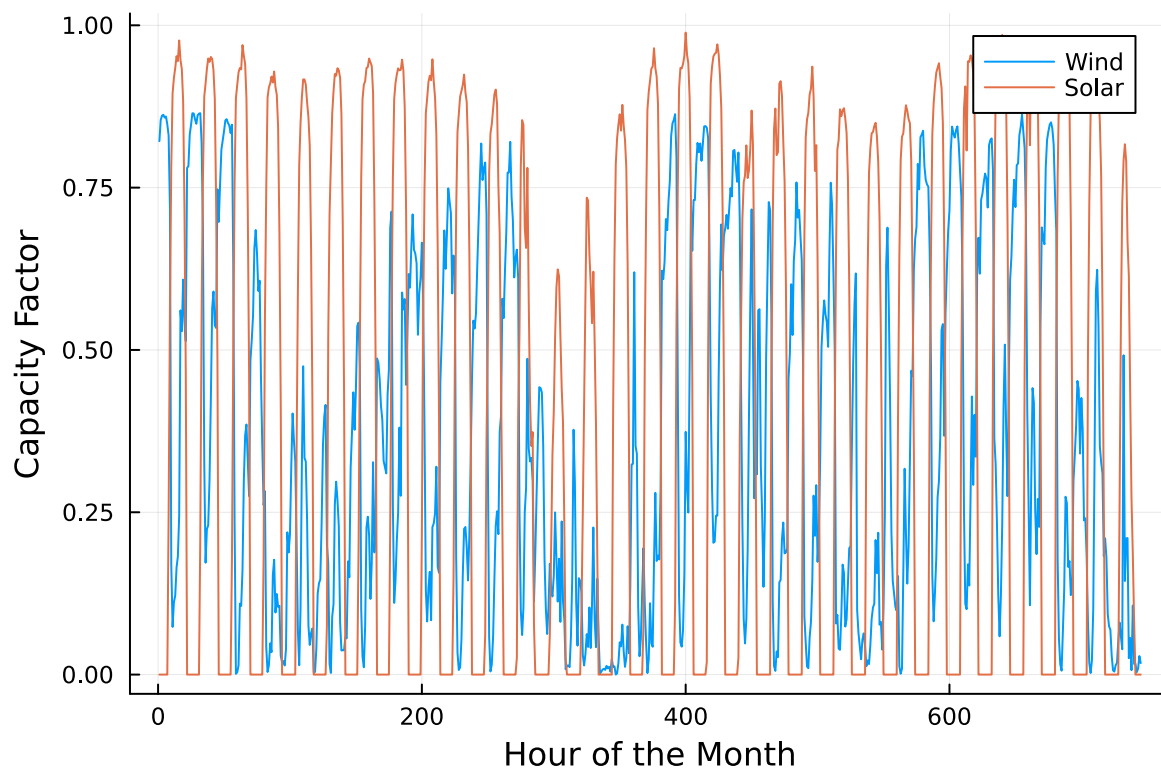
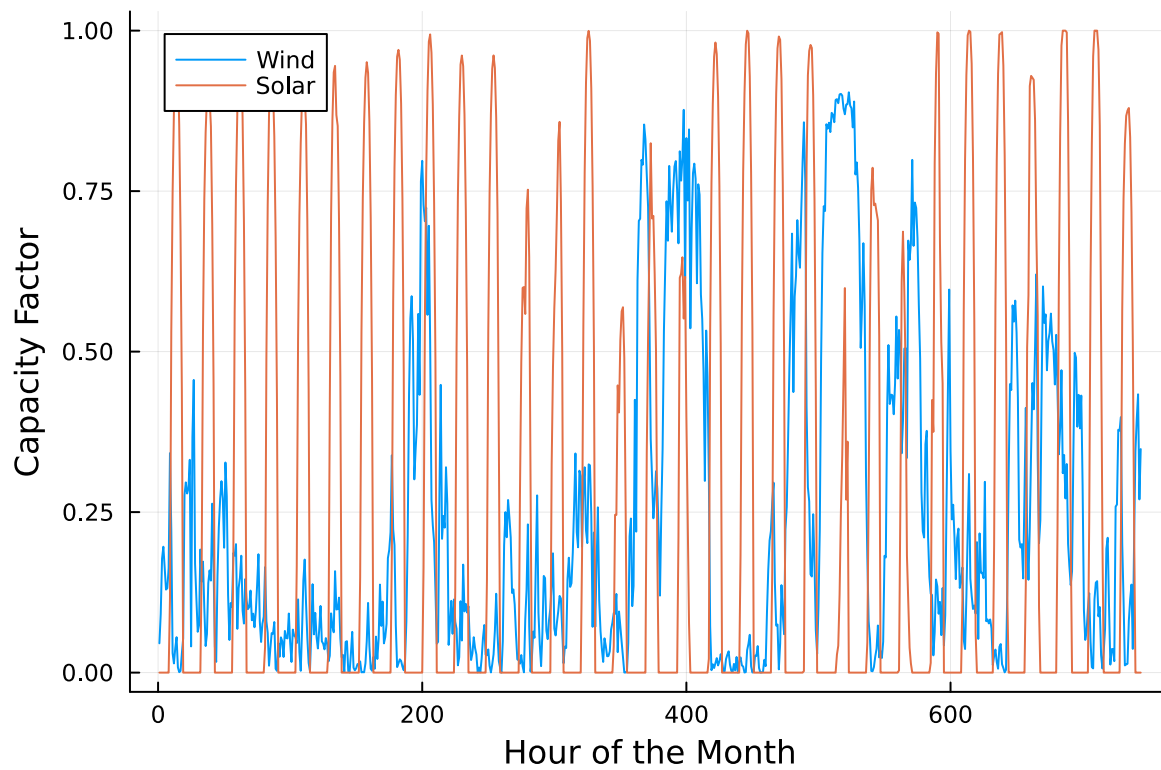
While coal, CCGT, and CT plants can generate at their full installed capacity, geothermal plants operate at maximum 85% capacity, and solar and wind available capacities vary by the hour depend on the expected meteorology. The utility will also penalize any non-served demand at a rate of \$10,000/MWh.

Problem 3.1

Formulate a linear program for this capacity expansion problem, including clear definitions of decision variable(s) (including units), objective function(s), and constraint(s) (make sure to explain functions and constraints with any needed derivations and explanations).

Problem 3.2

How much should the utility build of each type of generating plant? What will the total cost be? How much energy will be non-served?



Problem 3.3

What fraction of annual generation does each plant type produce? How does this compare to the breakdown of built capacity that you found in Problem 3.2? Do these results make sense given the demand and generator data?

Problem 3.5

What is the shadow price of CO₂? What is the value to the utility be of allowing it to emit an additional 1000 tCO₂/yr?

Significant Digits

Use `round(x; digits=n)` to report values to the appropriate precision! If your number is on a different order of magnitude and you want to round to a certain number of significant digits, you can use `round(x; sigdigits=n)`.

Getting Variable Output Values

`value.(x)` will report the values of a JuMP variable `x`, but it will return a special container which holds other information about `x` that is useful for JuMP. This means that you can't use this output directly for further calculations. To just extract the values, use `value.(x).data`.

Suppressing Model Command Output

The output of specifying model components (variable or constraints) can be quite large for this problem because of the number of time periods. If you end a cell with an `@variable` or `@constraint` command, I *highly* recommend suppressing output by adding a semi-colon after the last command, or you might find that your notebook crashes.

```
#Problem 3

gen_model = Model(HiGHS.Optimizer) #initialize model object

#establish constants

hours = 8760
emission_cap = 1500000
penalty = 10000
num_gen = 6
gen_names = gens.Plant #gen names from gens dataframe

#matrix to hold capacity values to be used later on for constraints
```

```

#this goes through every hour within the yearly data, 8760 hours,
#and for each hour sets values for the 6 generator types
#we know Geothermal runs at 0.85 capacity, and coal NG CCGT and NG CT
#can run at 1.0 capacity, and then for each hour we pull the hourly
#data from the cap factor dataframe and set that in our matrix
#to be utilized later
Cap_matrix = zeros(hours, num_gen)
for g in 1:6

    if gen_names[g] == "Geothermal"
        Cap_matrix[:, g] .= 0.85
    elseif gen_names[g] in ["Coal", "NG CCGT", "NG CT"]
        Cap_matrix[:, g] .= 1.0
    elseif gen_names[g] == "Wind"
        Cap_matrix[:, g] = cap_factor.Wind
    elseif gen_names[g] == "Solar"
        Cap_matrix[:, g] = cap_factor.Solar
    end

end

#establish decision variables
#decision variables will be the generator capacity (MW),
#the power generated by generator per hour (MWh), and
#the unserved load hours (MWh)

@variable(gen_model, Cap[1:6] >= 0)
@variable(gen_model, GenUse[1:6, 1:hours] >= 0)
@variable(gen_model, Unserved[1:hours] >= 0)

#establish objective function
#here we want to minimize the annual cost
#while still producing enough energy to meet all energy demands
#and emissions constraints

#to do so, we take the sum of the fixed costs multiplied by the generator capacity
#for each of the generator types, add it the sum of the variable costs to
#determine the MWh cost for each of the generator types, and then add the
#sum of all unserved load hours multiplied by the non-served demand penalty
#rate. By minimizing these three equations, we should get the minimized

```

```

#annual costs.

@objective(gen_model, Min, sum(gens.FixedCost[g] * Cap[g] for g = 1:num_gen) +
sum(gens.VarCost[g] * GenUse[g,t] for g = 1:num_gen, t in 1:hours)
+ sum(penalty * Unserved[t] for t in 1:hours))

#establish constraints

#hourly demand, all hours must be accounted for, either with generation
#or non-served which will be penalized

@constraint(gen_model, [t = 1:hours], (sum(GenUse[g, t] for g = 1:6) +
Unserved[t]) == demand.Demand[t])

#capacity limits per generator, this makes sure that generator usage
#is not greater than the capacity.

@constraint(gen_model, [g = 1:6, t = 1:hours], GenUse[g, t] <= Cap[g] *
Cap_matrix[t,g])

#constraint for the CO2 emissions cap, with shadow price included for problem 3.5,
#this multiplies emissions by generator use to ensure it is less than the provided
#emissions footprint.

@constraint(gen_model, co2_limit, sum(gens.Emissions[g] * GenUse[g, t]
for g in 1:6, t in 1:hours) <= emission_cap)

#optimize model
optimize!(gen_model)

#print results code

println("Total Cost", objective_value(gen_model))

#prints capacity for each generator type
println("Installed capacity (MW):")
for g = 1:6
    println("  ", gen_names[g], ": ", value(Cap[g]))
end

```

```

println("Unserved demand (MWh): ", (sum(value.(Unserved))))

#prints emissions by totaling emissions amount for each generator
#multiplied by each generator types usage hours
println("Total CO emissions (t): ",
        sum(gens.Emissions[g] * value(GenUse[g, t]) for g in 1:6, t in 1:hours))

#problem 3.3

#gets the sum of MWh for each generator type and then divides it by total
#generator MWh to determine fraction of MWh for each generator type

annual_gen = [sum(value.(GenUse[g, :])) for g in 1:6]
total_gen = sum(annual_gen)
frac_gen = annual_gen ./ total_gen

#print code for fraction produced by each generator type
for g = 1:6
    println("Plant: ", gens.Plant[g], ", Annual Generation (MWh): ", annual_gen[g],
            ", Fraction: ", round(frac_gen[g]*100, digits=2), "%")
end

#problem 3.5

#locates shadow price as established earlier for CO2
co2_sp = shadow_price(co2_limit)
println("Shadow price of CO constraint:", co2_sp, " per tCO ")

```

Running HiGHS 1.11.0 (git hash: 364c83a51e): Copyright (c) 2025 HiGHS under MIT licence terms
 LP has 61321 rows; 61326 cols; 188256 nonzeros

Coefficient ranges:

Matrix	[5e-05, 1e+00]
Cost	[2e+01, 4e+05]
Bound	[0e+00, 0e+00]
RHS	[1e+03, 2e+06]

Presolving model

56857 rows, 56862 cols, 179328 nonzeros 0s

Dependent equations search running on 8760 equations with time limit of 1000.00s

Dependent equations search removed 0 rows and 0 nonzeros in 0.00s (limit = 1000.00s)

56857 rows, 56862 cols, 179328 nonzeros 0s

Presolve : Reductions: rows 56857(-4464); columns 56862(-4464); elements 179328(-8928)

Solving the presolved LP
Using EKK dual simplex solver - serial

Iteration	Objective	Infeasibilities num(sum)
0	0.0000000000e+00	Pr: 8760(5.18855e+06) 0s
25125	6.7999737832e+08	Pr: 11013(9.12909e+06); Du: 0(1.0619e-07) 5s
31379	7.8276629217e+08	Pr: 1901(152998); Du: 0(1.34904e-07) 11s
32698	7.8535295894e+08	Pr: 0(0); Du: 0(2.91607e-11) 12s

Solving the original LP from the solution after postsolve
Model status : Optimal
Simplex iterations: 32698
Objective value : 7.8535295894e+08
P-D objective error : 1.3357583179e-14
HiGHS run time : 12.21
Total Cost 7.8535295894349e8
Installed capacity (MW):
Geothermal: 285.5681352306725
Coal: 0.0
NG CCGT: 1509.1728698942882
NG CT: 703.0618976373823
Wind: 2548.8879233664907
Solar: 2103.7493968268914
Unserved demand (MWh): 332.29071379259335
Total CO emissions (t): 1.5000000000000456e6
Plant: Geothermal, Annual Generation (MWh): 1.0969954555635913e6, Fraction: 6.7%
Plant: Coal, Annual Generation (MWh): 0.0, Fraction: 0.0%
Plant: NG CCGT, Annual Generation (MWh): 3.3227665594545435e6, Fraction: 20.3%
Plant: NG CT, Annual Generation (MWh): 129473.41715377511, Fraction: 0.79%
Plant: Wind, Annual Generation (MWh): 6.451743719237155e6, Fraction: 39.42%
Plant: Solar, Annual Generation (MWh): 5.366594457877142e6, Fraction: 32.79%
Shadow price of CO constraint: -182.77193609185866 per tCO

Problem 3.2: How much should the utility build of each type of generating plant? What will the total cost be? How much energy will be non-served?

The utility should build the following capacities for each type of generating plant

Geothermal, 285.57 MW Coal, 0.0 MW NG CCGT, 1,509.17 MW NG CT, 703.06 MW Wind, 2,548.89 MW Solar, 2,103.75 MW

The total cost will be approximately 785,000,000 dollars.

332.29 MWh will be non-served

Problem 3.3: What fraction of annual generation does each plant type produce? How does this compare to the breakdown of built capacity that you found in Problem 3.2? Do these results make sense given the demand and generator data?

Plant: Geothermal, Fraction: 6.7% Plant: Coal, Fraction: 0.0% Plant: NG CCGT, Fraction: 20.3% Plant: NG CT, Fraction: 0.79% Plant: Wind, Fraction: 39.42% Plant: Solar, Fraction: 32.79%

Compared to the breakdown of the built capacity for problem 3.2, these results seem to pair up well. There is a large presence in generation from both wind and solar as their variable costs and total emissions are 0. NG CCGT also has a notable percentage and would typically be used when demand is too great for just wind and solar as it is a relatively efficient method compared to NG CT and coal. Geothermal is considerably lower due to the limited capacity and NG CT is very low and is only used when absolutely necessary because while it can generate a lot of energy, it does have higher variable costs and emissions. The 0.0% fraction of coal also matches up with the result from 3.2

I would say these results do make sense given the demand and generator data as there is a large emphasis on the renewable energy methods which have no variable costs and no emissions. Then, there is the avoidance of coal, largely due to its emissions contributions, which makes sense and a “when-necessary usage” of NG CCGT and NG CT in even more dire situations.

Problem 3.5: What is the shadow price of CO₂? What is the value to the utility of allowing it to emit an additional 1000 tCO₂/yr?

The shadow price of CO₂ is -182.77 dollars per tCO₂

This means that if the utility were allowed to increase emissions by 1 ton of CO₂ per year, then the total cost would decrease by 182.77 dollars. Therefore, the value to the utility of allowing an additional 1000 tCO₂/year to be emitted would be a total cost decrease of 182,770 dollars.

References

OpenAI. (2024). ChatGPT (Sept 25 GPT-5 version) [Large language model]. <https://chat.openai.com>: I utilized ChatGPT to help specifically with understanding and implementing a lot of the Jump syntax, I did also utilize it for some julia-specific syntax questions, but it largely revolved around properly utilizing Jump as I struggled with some of the syntax and code nuances. All the comments, interpretations, coming up with constraints, etc. was done by me